Hypothesis and result

Objective: To formulate a hypothesis, describe an experiment and report its result using the language of scientific investigation.

The idea in two lines: A **hypothesis** is a possible explanation that we test; the **result** is what we actually observe. The result either confirms or rejects the hypothesis, and the language changes from uncertainty (may, might) to certainty (so, therefore).

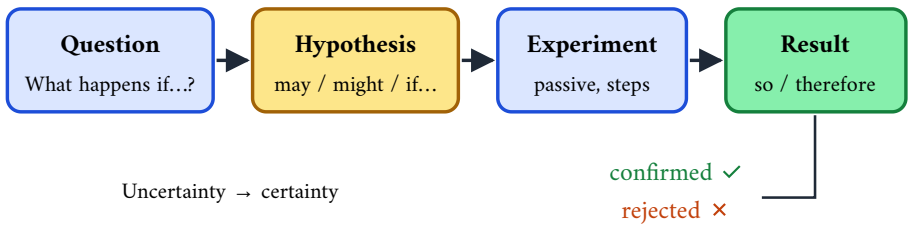


Figure 1: The steps of a scientific investigation, from a question to a result that confirms or rejects the hypothesis.

🔵 Rules and concepts

1. **Hypothesis with modals:** may / might / could + base form: *Light might help plants grow faster.*
2. **Hypothesis with if:** first conditional: *If we heat ice, it will melt.* For general scientific facts, use the zero conditional: *If you heat ice, it melts.*
3. **Expressions:** *We suppose / assume / expect that …; It is possible that …*
4. **Experiment:** described in the passive with sequencers: *Two plants were placed …*
5. **Result:** so, therefore, as a result, consequently; *The result shows / proves that …*
6. **Conclusion:** *The hypothesis is confirmed / rejected.*

🟢 Examples

1. Express a hypothesis: "Salt / make water boil at a higher temperature". **Answer:** Salt *might make* water boil at a higher temperature.
2. Report a result: "The plant in the dark turned yellow." **Answer:** The plant in the dark turned yellow; *therefore*, light is necessary for it to stay healthy.
   Explanation: observation + result connector + conclusion.
3. Zero conditional: "heat / metal / expand". **Answer:** If you heat metal, it expands.

🟠 Common BAC mistakes

- Stating a hypothesis as a certain fact (no **may** / **might**).
- Writing *might to grow* × instead of **might grow** ✓.
- Using *will* in the if clause.
- Confusing the **hypothesis** (before the test) with the **result** (after the test).

🟡 Quick review

- Hypothesis = may / might / could + verb.
- Scientific fact = if + present, present.
- Experiment = passive + sequencers.
- Result = so / therefore / the result shows that.
- End by confirming or rejecting the hypothesis.

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